



RRSI: Regularized Receptive Self-Improvement of Agent Harnesses

PAPER EXPLAINED

Stop overfitting your agent harness to benchmarks

RRSI: Regularized Recursive Self-Improvement of Agent Harnesses

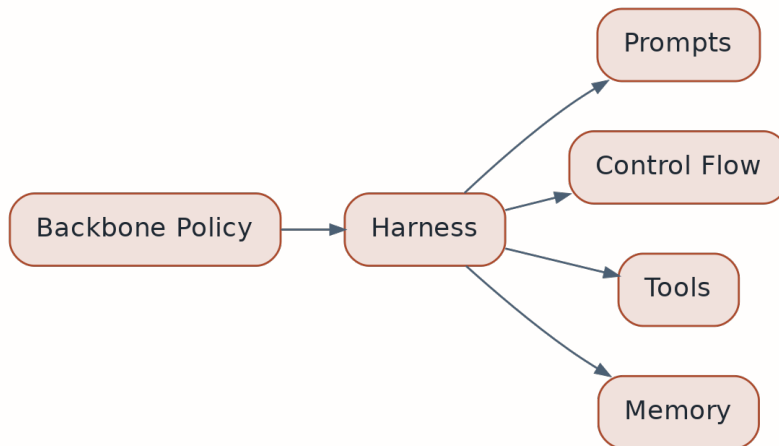
Peng Xia, Rujun Han, Zifeng Wang, Yanfei Chen et al.

<https://arxiv.org/abs/2609.24972>

TASK

Agent Harness Evolution Problem

- The paper defines an agent as a combination of a backbone policy and a harness managing prompts, control flow, tools, and memory.
- Task performance is the expected score on a task set, while policy-token cost is the expected number of tokens consumed.
- Harness evolution treats the harness as the optimization variable while keeping the backbone policy fixed.



Agent architecture: a fixed backbone policy optimized via harness components like prompts, control flow, tools, and memory.

Overfitting Risks in Iterative Evolution

- Iterative harness evolution uses finite feedback, creating adaptive overfitting where evolve-set performance fails to generalize to unseen tasks.
- Search encodes benchmark-specific patterns and promotes candidates favored by evaluation noise, accumulating complexity without improving the agent mechanism.
- Existing approaches rely on benchmark scores without generalization terms, so reported gains often do not survive changes in the evaluation suite.

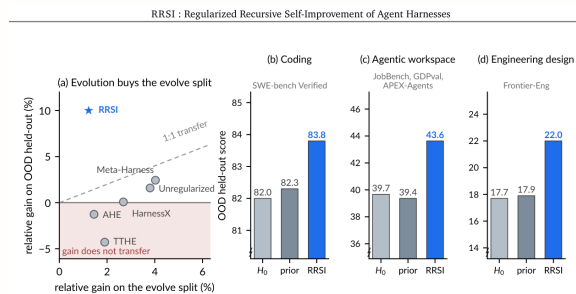
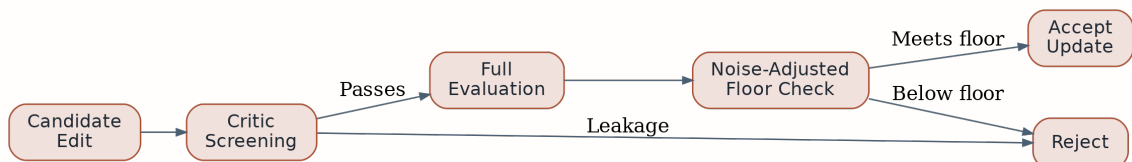


Figure 1 | Evolution overfits on the split it is scored on whereas RRSI generalizes the improvements. (a) Gains on the evolve split against gains out of distribution for the agentic workspace benchmark.

RRSI Regularization Framework Details

- RRSI anneals update capacity per round, uses evidence-aware credit assignment, and structures exploration.
- Leakage screening rejects candidates encoding task names or specific answers before full evaluation.
- Stability-aware acceptance uses a noise-adjusted floor to prevent selecting candidates due to stochastic variation.



RRSI regularization pipeline: screening for leakage, evaluation with a noise-adjusted stability floor.

Results, Gains, and Limitations

- RRSI improves six held-out splits by up to 4.7 points out of distribution using fewer policy tokens than unregularized evolution.
- The method gains 2.3 points on Harvey LAB and 1.8 points on SWE-bench Verified for in-distribution held-out tasks.
- Removing acceptance constraints raises the evolve-set score to 91.5 but drops out-of-distribution transfer from 43.6 to 41.0 and increases token cost.

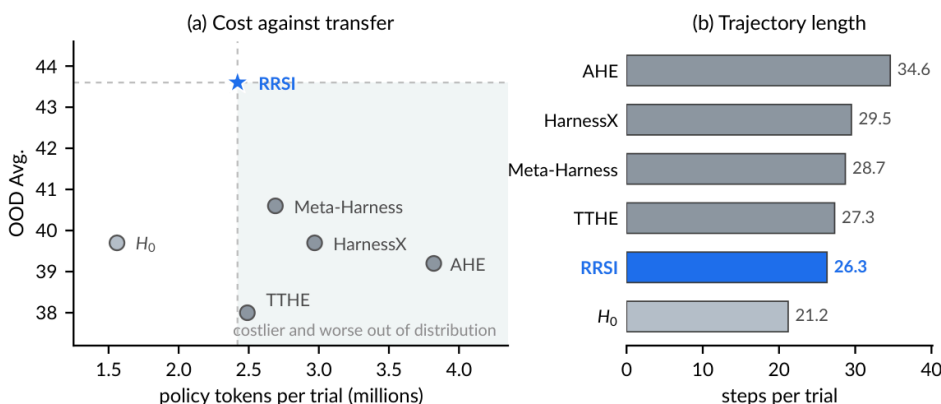


Figure 4 | Cost of the final harness of each arm, measured on the evolve split of the agentic workspace instance.